Math211
Discrete Mathematics

Mathematical Induction

Agenda

5.1 Mathematical Induction

Discrete Mathematics and Its Applications

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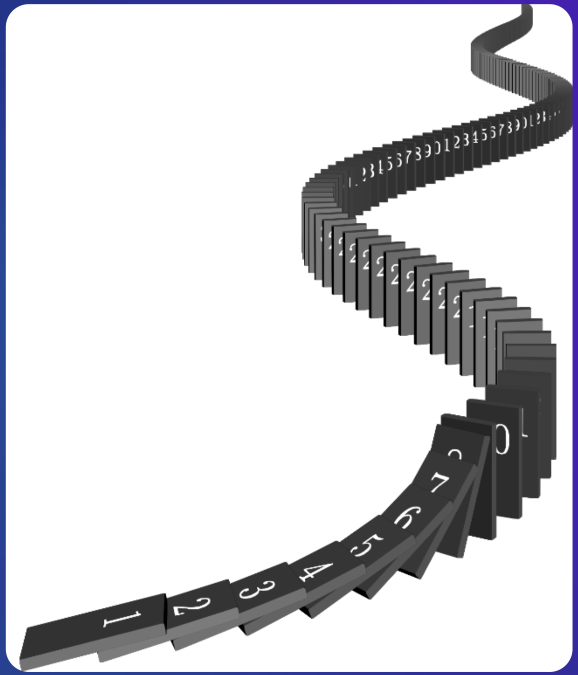
5.1. Mathematical Induction



- Proving Summation
- Proving Inequalities
- Proving Divisibility
- DeMorgans's Generalization

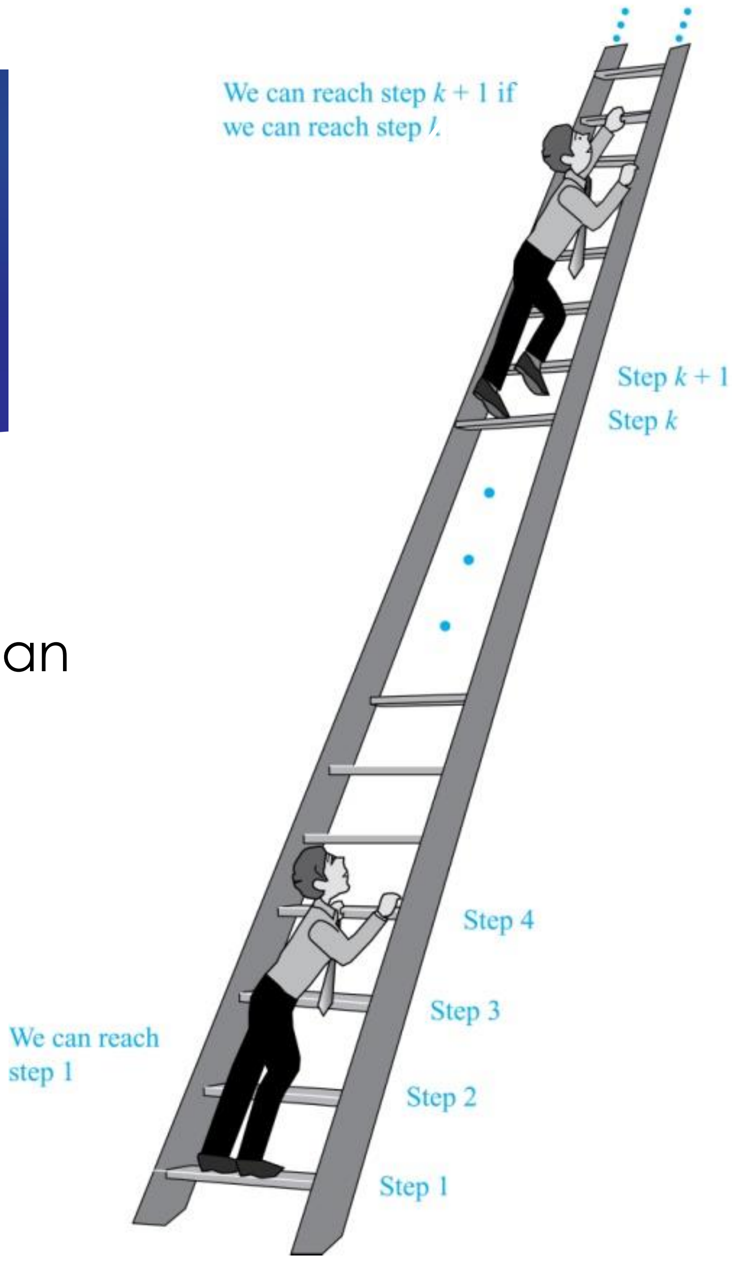
Climbing an Infinite Ladder

Suppose we have an infinite ladder:

1. We can reach the first rung of the ladder.
2. If we can reach a particular rung of the ladder, then we can reach the next rung.

- From (1), we can reach the first rung.
- Then by applying (2), we can reach the second rung.
- Applying (2) again, the third rung. And so on.
- We can apply (2) any number of times to reach any particular rung, no matter how high up.

This example motivates proof by mathematical induction.



Principle of Mathematical Induction

- S= 3 + 7 + 10 **(two addition operations)**
- S= 1 + 2 + 3 + 4 + … + 1000 **(1000 – 1 addition operation)**
- S= 1 + 2 + 3 + . . . +n = $\frac{n(n+1)}{2}$

  S= 1 + 2 + 3 + … +1000 = $\frac{1000\,(1000+1)}{2}$ **(Only three operations instead of 999 operations).**
- Mathematical induction is used to prove that the proposition function **P(n)=1 + 2 + 3 + . . . +n =** $\boldsymbol{\frac{n(n+1)}{2}}$ is true for all positive integers.

Principle of Mathematical Induction

To prove that $P(n)$ is true for all positive integers n, we complete these steps:

1. ***Basis Step:*** Show that $P(1)$ is true.
2. ***Inductive Step:*** Show that $P(k) \rightarrow P(k+1)$ is true for all positive integers k.

To complete the inductive step, assuming the *inductive hypothesis (IH)* that $P(k)$ holds for an arbitrary integer k, show that must $P(k+1)$ be true.

Climbing an Infinite Ladder Example:

1. **BASIS STEP:** By (1), we can reach rung 1.
2. **INDUCTIVE STEP:** Assume the inductive hypothesis that we can reach rung k. Then by (2), we can reach rung $k + 1$.

Hence, $P(k) \rightarrow P(k+1)$ is true for all positive integers k. We can reach every rung on the ladder.

Mathematical Induction

- It mainly deals with discrete math
  - Integers $\mathbb{Z}$ and positive integers $\mathbb{Z}^+$

T A P E

1. **T**est
2. **A**ssumption
3. **P**rove
4. **E**xplain

Basis step

Inductive step

Important Points About Using Mathematical Induction

- Mathematical induction can be expressed as the rule of inference

$$(P(1) \land \forall k\,(P(k) \rightarrow P(k+1))) \rightarrow \forall n\, P(n),$$

where the domain is the set of positive integers.

- In a proof by mathematical induction, we don't assume that $P(k)$ is true for all positive integers! We show that if we assume that $P(k)$ is true, then $P(k+1)$ must also be true.
- Proofs by mathematical induction do not always start at the integer 1. In such a case, the basis step begins at a starting point b where b is an integer.

Remembering How Mathematical Induction Works

Consider an infinite sequence of dominoes, labeled $1,2,3, \ldots,$ where each domino is standing.

Let $P(n)$ be the proposition that the n^{th} domino is knocked over.

We know that the first domino is knocked down, i.e., $P(1)$ is true.

We also know that if whenever the k^{th} domino is knocked over, it knocks over the $(k+1)^{st}$ domino, i.e, $P(k) \rightarrow P(k+1)$ is true for all positive integers k.

Hence, all dominos are knocked over.

$P(n)$ is true for all positive integers n.

Important Points About Using Mathematical Induction

- To prove that the propositional function P(n) is true for all numbers in the given domain **[The set of integers or positive integers]**, the following steps are done:

1. **Basis Step:** Show that P(n) is true at the first number in the domain [The first step is 1 and 0 for the set of the positive and non-negative integers, respectively].
2. **Inductive Step:** Assume that the inductive hypothesis is true [P(k)] and show that P[k+1] is true.

Important Points About Using Mathematical Induction

- **Generally, prove that P(K+1) is true using P(K). This can be done in two ways:**

1. Start from P(K) and modify the LHS and RHS of the equation by **adding/subtracting/multiplying/dividing** another term to both sides till the LHS of P(K) reaches (equals) the LHS of P(K+1). Then, simplify the LHS of the modified P(k) till its value is equal to the RHS of P(K+1).
2. Start from the LHS of P(K+1) and simplify the LHS term using P(K) till its value reaches (equal) the RHS of P(K+1)

- The first and second methods can be applied in both equality and inequality proofs, while only the second method can be applied in the divisibility proof.

Proving Summation

Proving Summation Formula: Example 1

- Show that 1+2+3+ . . . +n = $\frac{n(n+1)}{2}$ where n is $\mathbb{Z}^+$

- Let $P(n)$ be the proposition that 1+2+3+ . . . +n = $\frac{n(n+1)}{2}$

1. **Basis step:** Prove P(1) is true
2. **Inductive step:** Assume P(k) is true, then prove that P(k+1) is true
   - Inductive Hypothesis P(k): 1+2 + 3 + . . . k = $\frac{k(k+1)}{2}$
   - Prove that P(k+1) is true: 1+2+3+…+k+k+1 = $\frac{(k+1)(k+2)}{2}$

Method 1

IH P(K): $1+2+3+\ldots k = \frac{k(k+1)}{2}$

Prove P(K+1): $1+2+3+\ldots+k+(k+1) = \frac{(k+1)(k+2)}{2}$

1. **Basis Step:** Prove P(1) is true

- P(1): $1 = \frac{1(1+1)}{2} = 1$, So P(1) is true

2. **Inductive Step:** Assume P(k) is true, then prove that P(k+1) is true

- $1+2+3+\ldots k = \frac{k(k+1)}{2}$ (by assumption) [add (k+1) to both sides]
- $1+2+3+\ldots k + (k+1) = \frac{k(k+1)}{2} + (k+1)$

$$= \frac{k(k+1)}{2} + \frac{2(k+1)}{2} = \frac{k(k+1)+2(k+1)}{2}$$

$$= \frac{(k+1)(k+2)}{2}$$

LHS of P(K+1) = RHS of P(K+1)

- Therefore, $1+2+3+\ldots+n = \frac{n(n+1)}{2}$ where n is $\mathbb{Z}+$

Method 2

IH P(K): $1+2+3+\ldots k = \frac{k(k+1)}{2}$

Prove P(K+1): $1+2+3+\ldots+k+(k+1) = \frac{(k+1)(k+2)}{2}$

1. **Basis Step:** Prove P(1) is true

- P(1): $1 = \frac{1(1+1)}{2} = 1$, So P(1) is true

2. **Inductive Step:** Assume P(k) is true, then prove that P(k+1) is true

- **LHS of P(K+1): 1+2 + 3 + . . . k** + k+1 = $\frac{\boldsymbol{k(k+1)}}{\mathbf{2}}$ + k+1

$$= \frac{k(k+1)}{2} + \frac{2(k+1)}{2} = \frac{k(k+1)+2(k+1)}{2}$$

$$= \frac{(k+1)(k+2)}{2}$$

$$= \frac{(k+1)(k+2)}{2}$$

= RHS of P(K+1)

- Therefore, $1+2+3+\ldots+n = \frac{n(n+1)}{2}$ where n is $\mathbb{Z}+$

Proving Summation Formula: Example 2

- Use mathematical induction to show that

$1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ for all non-negative integers n.

- Let $P(n)$ be the proposition that $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$

1. **Basis Step:** Prove P(0) is true

2. **Inductive step:** Assume P(k) is true and Prove that P(k+1) is true.
   - Inductive Hypothesis P(k): $1 + 2 + 2^2 + \dots + 2^k = 2^{k+1} - 1$
   - Must show P(k+1) is true: $1 + 2 + 2^2 + \dots + 2^k + 2^{k+1} = 2^{k+2} - 1$

Method 1

$1 + 2 + 2^2 + ... + 2^n = 2^{n+1} - 1$

P(K): $1 + 2 + 2^2 + ... + 2^k = 2^{k+1} - 1$
P(K+1): $1 + 2 + 2^2 + ... + 2^k + 2^{k+1} = 2^{k+2} - 1$

1. **Basis Step:** Prove P(0) is true

   P(0): $2^0 = 2^{0+1} - 1$

   $1 = 1$ ✓

2. **Inductive Step:** Assume P(k) is true, then prove that P(k+1) is true

- $1 + 2 + 2^2 + ... + 2^k = 2^{k+1} - 1$ (using IH) [add 2^{k+1} to both sides]
- $1 + 2 + 2^2 + ... + 2^k + 2^{k+1} = 2^{k+1} - 1 + 2^{k+1}$

$$= 2(2^{k+1}) - 1$$

$$= 2^{k+2} - 1$$

LHS of P(K+1) = RHS of P(K+1)

- Therefore, $1 + 2 + 2^2 + ... + 2^n = 2^{n+1} - 1$ for all non-negative integers n.

Method 2

$1 + 2 + 2^2 + \ldots + 2^n = 2^{n+1} - 1$

$P(K): 1 + 2 + 2^2 + \ldots + 2^k = 2^{k+1} - 1$
$P(K+1): 1 + 2 + 2^2 + \ldots + 2^k + 2^{k+1} = 2^{k+2} - 1$

1. **Basis Step:** Prove P(0) is true

   $P(0): 2^0 = 2^{0+1} - 1$

   $1 = 1$ ✓

2. **Inductive Step:** Assume P(k) is true, then prove that P(k+1) is true

- **LHS of P(K+1):** $1 + 2 + 2^2 + \ldots + 2^k + 2^{k+1}$

  $= 2^{k+1} - 1 + 2^{k+1} = 2(2^{k+1}) - 1 = 2^{k+2} - 1$

  = RHS of P(K+1)

- Therefore, $1 + 2 + 2^2 + \ldots + 2^n = 2^{n+1} - 1$ for all non-negative integers n.

Example 3

- Prove $1+3+5+ \ldots +(2n-1) = n^2 \quad n \in Z^+$

- Let P(n) be the proposition that $1+3+5+ \ldots +(2n-1) = n^2$

1. **Basis step:** Prove P(1) is true

2. **Inductive step:** Assume P(k) is true, then prove that P(k+1) is true
   - Inductive Hypothesis P(k): $1+3+5+ \ldots +(2k-1) = k^2$
   - Prove that P(k+1) is true: $1+3+5+ \ldots +(2k-1) + (2k+1) = (k+1)^2$

IH P(K): $1+3+5+ \ldots +(2k-1) = k^2$
Prove P(K+1): $1+3+5+ \ldots +(2k-1) + (2k+1) = (k+1)^2$

1. **Basis step:** Prove P(1) is true
   $1 = (1)^2$
   $1 = 1$

2. **Inductive Step:** Assume P(k) is true, then prove that P(k+1) is true [add (2k+1) to both sides] (by Inductive hypothesis)

- $1+3+5+ \ldots +(2k-1) = k^2$

  $1+3+5+ \ldots +(2k-1) + (2k+1) = k^2 + (2k+1)$

  $= (k+1)^2$

LHS of P(K+1) = RHS of P(K+1)

- Therefore, $1+3+5+ \ldots +(2n-1) = n^2 \quad n \in Z+$

IH P(K): $1+3+5+ \dots +(2k-1) = k^2$
Prove P(K+1): $1+3+5+ \dots +(2k-1) + (2k+1) = (k+1)^2$

1. **Basis step:** Prove P(1) is true
   $1 = (1)^2$
   $1 = 1$

2. **Inductive Step:** Assume P(k) is true, then prove that P(k+1) is true

- **LHS of P(K+1):** $1+3+5+ \dots +(2k-1) + (2k+1)$

  $= K^2 + 2k + 1$

  $= (K+1)^2$ = RHS of P(K+1)

  = RHS of P(K+1)

- Therefore, $1+3+5+ \dots +(2n-1) = n^2 \quad n \in Z+$

Proving Inequalities

Proving Inequalities: Example 1

Use mathematical induction to prove that $n < 2^n$ for all positive integers n.

Solution: Let $P(n)$ be the proposition that $n < 2^n$.

1. **BASIS STEP:** *Prove P(1) is true,* $P(1)$ is true since $1 < 2^1 = 2$.
2. **INDUCTIVE STEP:** Assume P(k) is true, then prove that P(k+1) is true
   - Assume $P(k)$ is true, i.e., $k < 2^k$, for an arbitrary positive integer k.
   - Prove that $P(k + 1)$:$k + 1 < 2^{k+1}$ is true.
   - Since by the inductive hypothesis, $k < 2^k$, it follows that:

     $k + 1 < 2^k + 1 < 2^k + 2^k < 2^k\ (1+1) < 2 \cdot 2^k < 2^{k+1}$
   - LHS of P(k+1) < RHS of P(K+1)
   - Therefore n < 2n holds for all positive integers n.

Method 1

Proving Inequalities: Example 1

Solution: Let $P(n)$ be the proposition that $n < 2^n$.

1. **BASIS STEP:** *Prove P(1) is true,* $P(1)$ is true since $1 < 2^1 = 2$.
2. **INDUCTIVE STEP:** Assume P(k) is true, then prove that P(k+1) is true
   - Assume $P(k)$ holds, i.e., $k < 2^k$, for an arbitrary positive integer k.
   - Prove that $P(k + 1)$ holds, i.e., $k + 1 < 2^{k+1}$.
   - LHS of P(k+1) $= k + 1 < 2^k + 1 < 2^k + 1 < 2^k + 2^k < 2.\ 2^k < 2^{k+1}$
   - LHS of P(k+1) < RHS of P(K+1)
   - Therefore $n < 2^n$ holds for all positive integers n.

Method 2

Proving Inequalities: Example 2

Note that here the basis step is $P(4)$, since $P(0)$, $P(1)$, $P(2)$, and $P(3)$ are all false.

Use mathematical induction to prove that $2^n < n!$, for every integer $n \geq 4$.

Solution: Let $P(n)$ be the proposition that $2^n < n!$

1. **BASIS STEP:** *Prove P(4) is true,* $P(4)$ is true since $2^4 = 16 < 4! = 24$
2. **INDUCTIVE STEP:** Assume P(k) is true, then prove that P(k+1) is true
   - Assume P(k) holds, i.e., $2^k < k!$ for an arbitrary integer $k \geq 4$
   - Prove that $P(k + 1)$ is true i.e., $2^{k+1} < (k+1)!$ [LHS of P(k+1) < RHS of P(K+1)]
   - $2^k\, 2 < k!\,.\,2$ [multiply (2) to both sides]

     $< k!\,(k+1)$ $\quad \boxed{\because k \geq 4 \quad \therefore 2 < k}$

     $< (k + 1)!$
   - LHS of P(k+1) < RHS of P(K+1)
   - Therefore, $2^n < n!$ holds, for every integer $n \geq 4$.

Method 1

Proving Inequalities: Example 2

Solution: Let $P(n)$ be the proposition that $2^n < n!$

1. **BASIS STEP:** *Prove P(4) is true,* $P(4)$ is true since $2^4 = 16 < 4! = 24$
2. **INDUCTIVE STEP:** Assume P(k) is true, then prove that P(k+1) is true
   - Assume P(k) holds, i.e., $2^k < k!$ for an arbitrary integer $k \geq 4$
   - Prove that $P(k+1)$ i.e., $2^{k+1} < (k+1)!$ holds:

$$\begin{aligned} \text{LHS of } P(k+1) = 2^{k+1} &= 2 \cdot 2^k \\ &< 2 \cdot k! \qquad \textit{(by the inductive hypothesis)} \\ &< (k+1)k! \qquad \boxed{\because k \geq 4 \quad \therefore 2 < k} \\ &< (k+1)! \end{aligned}$$

   - LHS of P(k+1) < RHS of P(K+1)
   - Therefore, $2^n < n!$ holds, for every integer $n \geq 4$.

Method 2

Proving Divisibility

Proving Divisibility: Example 1

Use mathematical induction to prove that $n^3 - n$ is divisible by 3, for every positive integer n $\mathbb{Z}^+$.

Solution: Let $P(n)$ be the proposition that $n^3 - n$ is divisible by 3.

1. **BASIS STEP:** *Prove that P(1) is true*
   - P(1) is true since $1^3 - 1 = 0$, which is divisible by 3.
2. **INDUCTIVE STEP:** Assume P(k) is true, and prove that P(k+1) is true
   - Assume $P(k)$ is true, i.e., $k^3 - k$ is divisible by 3, for an arbitrary positive integer k.
   - Show that $P(k+1)$ is true, i.e., $(k+1)^3 - (k+1)$ is divisible by 3, for an arbitrary positive integer k.

Proving Divisibility: Example 1

$k^3 - k = 3 \times P$ where k, P $\in \mathbb{Z}^+$
$(K+1)^3 - k = 3 \times$ Q where Q $\in \mathbb{Z}^+$

Assume P(k) is true, i.e., $k^3 - k$ is divisible by 3, for an arbitrary positive integer k.
Show that P(k + 1)is true, $(k + 1)^3 - (k + 1)$ is divisible by 3

- $(k + 1)^3 - (k + 1) = (k^3 + 3k^2 + 3k + 1) - (k + 1)$

  $= (k^3 - k) + 3(k^2 + k)$

- By the inductive hypothesis, the first term $(k^3 - k)$ is divisible by 3 and the second term is divisible by 3 since it is an integer multiplied by 3. Thus, $(k + 1)^3 - (k + 1)$ is divisible by 3.
- $(k^3 - k) + 3(k^2 + k) = 3P + 3\ (k^2 + k) = 3(P + k^2 + k)$, Thus, $(k + 1)^3 - (k + 1)$ is divisible by 3.
- Therefore, $n^3 - n$ is divisible by 3, for every positive integer n.

Proving Divisibility: Example 2

Use mathematical induction to prove $7^{n+2} + 8^{2n+1}$ is divisible by 57 for all non-negative integers n.

- Let P(n) denote that $7^{n+2} + 8^{2n+1}$ is divisible by 57

1. Basis Step: Show P(0) is true
   - $7^{0+2} + 8^{2(0)+1} = 7^2 + 8^1 = 49 + 8 = 57$
2. Inductive Step: Assume P(k) is true, prove P(k+1) is true
   - IH: $7^{k+2} + 8^{2k+1}$ is divisible by 57
   - Show: $7^{(k+1)+2} + 8^{2(k+1)+1}$ is divisible by 57

Proving Divisibility: Example 2

$7^{k+2} + 8^{2k+1}$ = $57 \times P$ where k, P ϵ $\mathbb{Z}^{+}$
$7^{(k+1)+2} + 8^{2(k+1)+1}$ = $57 \times$ Q where Q ϵ $\mathbb{Z}^{+}$

IH: $7^{k+2} + 8^{2k+1}$ is divisible by 57
Show: $7^{(k+1)+2} + 8^{2(k+1)+1}$ is divisible by 57

- $$\begin{aligned} 7^{(k+1)+2} + 8^{2(k+1)+1} &= 7.\,7^{k+2} + 8^{2}\,.\,8^{2k+1} \\ &= 7.\,7^{k+2} + 64\,.\,8^{2k+1} \\ &= 7.\,(7^{k+2} + 8^{2k+1}) + 57\,.\,8^{2k+1} \end{aligned}$$
- By the inductive hypothesis, the first term $7^{k+2} + 8^{2k+1}$ is divisible by 57 and the second term is divisible by 57 since it is an integer multiplied by 57. Thus, $7^{(k+1)+2} + 8^{2(k+1)+1}$ is divisible by 57.
- $7.\,(7^{k+2} + 8^{2k+1}) + 57\,.\,8^{2k+1} = 57(7 + P + 8^{2k+1})$, thus $7^{(k+1)+2} + 8^{2(k+1)+1}$ can be divisible by 57.
- Therefore, $7^{(k+1)+2} + 8^{2(k+1)+1}$ is divisible by 57, all non-negative integers n.

Review Questions

Problem 1

Prove 3+6+9+12+ . . . +3n = $\frac{3n(n+1)}{2}$ **for n positive integer numbers**

- Assume P(n) is the proposition that **3+6+9+12+ . . . +3n =** $\frac{3n(n+1)}{2}$

1. Basis Step: Prove P(1) is true

   $3(1) = \frac{3(1)(1+1)}{2}$

   3 = 3 ✓

2. Inductive Step: Assume P(K) is true and Prove that P(K+1) is true

   P(k)= 3+6+9+12+ . . . +3k = $\frac{3k(k+1)}{2}$

   P(k+1)= 3+6+9+12+ . . . + 3k + 3(k+1) = $\frac{3(k+1)(k+2)}{2}$

Problem 1: Solution

- $3+6+9+12+ \ldots +3k + 3(k+1) = \frac{3k(k+1)}{2} + 3(k+1) \times \frac{2}{2}$ (by IH)

$$= \frac{3k(k+1)+6(k+1)}{2} = \frac{3(k+1)(k+2)}{2}$$

LHS of P(K+1) = RHS of P(K+1)

- **Therefore, $3+6+9+12+ \ldots +3n = \frac{3n(n+1)}{2}$ for n positive integer numbers**

Problem 2

- **Use mathematical induction to show that for all positive integers n, $1 + 2 + 2^2 + ... + 2^{n-1} = 2^n – 1$**

- Assume P(n) is the proposition that $1 + 2 + 2^2 + ... + 2^{n-1} = 2^n – 1$

1. Basis step: Prove P(1) is true

   $P(1): 2^{1-1} = 2^1 – 1$

   $1 = 1$ ✓

2. Inductive Step: Assume P(K) is true and Prove that P(K+1) is true

   - $P(K): 1 + 2 + 2^2 + ... + 2^{k-1} = 2^k – 1$
   - $P(K+1)=1 + 2 + 2^2 + ... + 2^{k-1} + 2^{k+1-1} = 2^{k+1} – 1$

Problem 2: Solution

- LHS of P(K+1)=$1 + 2 + 2^2 + . . . + 2^{k-1} + 2^{k+1-1}$

  $= (1 + 2 + 2^2 + . . . + 2^k) + 2^{k+1-1}$

  $= (2^k – 1) + 2^k$ **(by assumption)**

  $= 2.2^k - 1$

  $= 2^{k+1} – 1$

  LHS of P(K+1) = RHS of P(K+1)

- **Therefore, $1 + 2 + 2^2 + . . . + 2^{n-1} = 2^n – 1$, for all positive integers n**

Problem 3: Divisibility

IH: $k^3 + 3k^2 + 2k = 6P$

Prove: $(k+1)^3 + 3(k+1)^2 + 2(k+1) = 6Q$

Use mathematical induction to prove that $n^3 + 3n^2 + 2n$ is divisible by 6, for every positive integer n $\mathbb{Z}+$.

- Assume P(n) is the proposition that $n^3 + 3n^2 + 2n$ is divisible by 6

1. Basis Step: Prove P(1) is true
   - $1^3 + 3(1)^2 + 2(1) = 1+3+2 = 6$ *(True for n = 1)*

2. Inductive Step: Assume P(K) is true and Prove that P(K+1) is true
   - $P(K) = k^3 + 3(k)^2 + 2(k) = 6P$ where $k, P \in \mathbb{Z}+$
   - $P(K+1) = (k+1)^3 + 3(k+1)^2 + 2(k+1) = 6Q$ where $Q \in \mathbb{Z}+$

Problem 3: Solution

IH: $k^3 + 3k^2 + 2k = 6P$
Prove: $(k+1)^3 + 3(k+1)^2 + 2(k+1) = 6Q$

- L.H.S. of P(K+1) = $(k+1)^3 + 3(k+1)^2 + 2(k+1)$
  $= k^3 + 3k^2 + 3k + 1 + 3k^2 + 6k + 3 + 2k + 2$
  $= (k^3 + 3k^2 + 2k) + 3k + 1 + 3k^2 + 6k + 3 + 2$
  $= 6P + 3k^2 + 9k + 6$ (by assumption)
  $= (6P + 6) + (3k^2 + 9k)$
  $= (6P + 6) + 3k(k + 3)$ => but $k \in \mathbb{Z}^+$, hence either k or (k+3) must be even
  $= (6P + 6) + 3*2R$ where $2R = k(k + 3)$
  $= 6(P+1+R)$ => which is divisible by 6
- *Therefore, $n^3 + 3n^2 + 2n$ is divisible by 6, for every positive integer n* $\mathbb{Z}^+$.

Problem 4

$5^k + 3 = 4 \times P$ where k, P $\epsilon\ \mathbb{Z}^+$
$5^{k+1} + 3 = 4 \times Q$ where Q $\epsilon\ \mathbb{Z}^+$

Prove that $5^n + 3$ is divisible by 4, for n $\mathbb{Z}^+$.

- Assume P(n) is the proposition that $5^n + 3$ is divisible by 4, for n $\mathbb{Z}^+$.

1. Basis Step: Prove P(1) is true
   - $5^1 + 3 = 5+3 = 8 \therefore$ true for n=1
2. Inductive Step: Assume P(K) is true and Prove that P(K+1) is true
   - $P(K)= 5^k + 3 = 4 \times P$ where k, P $\epsilon\ \mathbb{Z}^+$
   - $P(K+1)= 5^{k+1} + 3 = 4 \times Q$ where Q $\epsilon\ \mathbb{Z}^+$

Problem 4: Solution

$5^k + 3 = 4 \times P$ where k, P $\epsilon\ \mathbb{Z}^+$
$5^{k+1} + 3 = 4 \times Q$ where Q $\epsilon\ \mathbb{Z}^+$

- LHS of P(K+1) = $5^{k+1} + 3$

  $= 5^{k+1} + 3$

  $= 5 . 5^k + 3$

  $= 4.5^k + 5^k + 3$

  $= 4.5^k + (5^k + 3)$ where $(5^k + 3)$ is divisible by 4 (by assumption)

  $= 4.5^k + 4P$

  $= 4(5^k + P)$

  $= 4$ Q where Q $= (5^k + P) =$ R.H.S. of P(K+1)
- Therefore, $5^n + 3$ is divisible by 4, for $n\ \mathbb{Z}^+$.

References

- Discrete Mathematics and Its Applications, Kenneth H. Rosen
- Michael P. Frank Slides

Supplementary Material

- Discrete Math - 5.1.1 Proof Using Mathematical Induction: Summation
- Discrete Math - 5.1.2 Proof Using Mathematical Induction: Inequality
- Discrete Math - 5.1.3 Proof Using Mathematical Induction: Divisibility
- Mathematical Induction - Proving Divisibility
  - https://www.youtube.com/watch?v=IdTaA6iz3Mo
  - https://www.youtube.com/watch?v=aG9gE6pjqAI

See you next lecture